Cover Letter to the Editor-in-Chief

2026-10-27

Date: October 27, 2026

To:
The Editors-in-Chief
Computers & Geosciences
Elsevier Science

Subject: Submission of original research paper titled *"A Reproducible Computational Framework for Decoupled Wildfire Ignition and Propagation Benchmarking in the Gran Chaco (2018–2025)"*

Dear Editors-in-Chief,

Please find enclosed our original research manuscript entitled **"A Reproducible Computational Framework for Decoupled Wildfire Ignition and Propagation Benchmarking in the Gran Chaco (2018–2025): D2 Targets, M3/M2 Inference and an Open Benchmark Against Canadian FWI and INPE"**, which we submit for consideration as a **Research Paper** in *Computers & Geosciences*.

Motivation and Core Scientific Novelty

Operational wildfire forecasting in geoscience machine learning frequently suffers from two pervasive methodological pathologies: 1. **Spatial Shortcut Learning:** When models include autoregressive spatial lags without isolating causal mechanisms, gradient-boosted ensembles learn trivial spatial persistence heuristics (*"fire will persist where thermal anomalies already exist"*), resulting in severe false confidence while remaining incapable of anticipating new, spontaneous, or anthropic primary ignitions. 2. **Failure of Canonical Boreal Indices in the Subtropics:** Widely deployed operational frameworks—such as the Canadian Forest Fire Weather Index (FWI) used by EFFIS/GWIS and the CPTEC/INPE Risco de Fogo—consistently fail or invert hazard rankings in water-limited subtropical biomes (e.g., the South American Gran Chaco), where vapor pressure deficit (VPD) and non-linear fuel-curing dynamics govern fire risk.

To address these challenges, we introduce `Incendios_PY`, an open-source computational framework for reproducible decoupled wildfire modelling across $400,006$ **1-km^2 grid cells** over an 8-year multi-annual observation baseline (2018–2025). This horizon is anchored to the multi-sensor VIIRS 375 m constellation (Suomi-NPP, NOAA-20 and NOAA-21), which reduces single-sensor sampling bias across the modeling phases. We analytically decouple the operational task into two distinct models: (i) **Multiscale Socio-Ecological Hybrid Ignition** ($\mathcal{M}_3$), guided by physics-based monotonic constraints, ecotone edge gradients, and static anthropogenic priors at 10-km and 25-km scales; and (ii) **Tactical Propagation**, driven by a causal spatial contagion kernel at lag $t-1$.

Why this Manuscript Fits *Computers & Geosciences*

Our work sits precisely at the convergence of **Computer Science Applications** and **Earth and Planetary Sciences**. The computational contribution is the D2 target operator, the gated M3/M2 inference policy and a reusable evaluation harness, released as documented software. The learner is replaceable; the novelty lies in the computational contract.

- **High-Performance Zero-Copy Computational Pipeline:** By integrating Apache Arrow, DuckDB, and Polars, our computational engine processes the entire national territory ($400,006$ km^2) in ~1.57 **seconds** on a standard ARM64 machine (~2.18 GB peak RAM), demonstrating that national early-warning systems do not require prohibitive cloud infrastructure.
- **Physics-Guided Machine Learning (PGML):** We enforce structural monotonic constraints within LightGBM (+1 for atmospheric dryness and wind pressure, −1 for antecedent rainfall and distance), mathematically preventing counterintuitive artifacts.
- **Spatial and Temporal Control:** We use the **D2 target definition** (no t-1 fire within 25 km; no same-cell fire in 10 days) and multi-year out-of-time test partitions (2024–2025). We also implement a **50-km spatial block holdout with a 20-km buffer**. Priors are causal (monthly cutoff, rolling 3-year window).
- **Software Engineering and Reproducibility Standards:** The framework is packaged with a complete automated test suite of **76 unit tests** (`tests/`) running in ≈ 1 minute, verifying algebraic invariants, physical monotonicity, conformal coverage, and spatial KD-Tree queries. All code and workflows are versioned from raw ingestion (`scripts/00_...`) to release and validation outputs (`scripts/34_...`) under the permissive MIT License.
- **Quantitative International Benchmarking:** In retrospective temporal testing over $1,813,966$ D2 cell-days during 2024–2025, our hybrid ignition layer $\mathcal{M}_3$ achieves a ROC-AUC of 0.8021 in the Gran Chaco and 0.7534 in the Oriental Region, versus 0.5173 / 0.5564 for the full Canadian FWI and 0.5389 / 0.4990 for the INPE. Case-control $F_{0.5}$ precision is 32.82 % / 30.13 %; after correction to population prevalence (~0.015 % / 0.041 %), operational precision is 0.082 % / 0.198 %. Tactical propagation achieves ROC-AUC 0.9039 (Chaco) and population precision 9.67 %. The 2023 validation selected

M3 for temporal nowcasting and M2 as the operational fallback; M1 was rejected as a fallback. These corrected metrics are the honest operational benchmark.

Mandatory Declarations

- This manuscript has not been published previously, is not under consideration for publication elsewhere, and its publication has been approved by the author.
- All source code and reproduction scripts are openly hosted on GitHub (`https://github.com/sanchezmarti code/Incendios_PY`). Processed data and trained models are documented with SHA-256 hashes in `fuentes/MANIFIESTO_DATOS_v3.csv`, available in the v2.0.0 GitHub release, and indexed by the code and data concept DOIs (10.5281/zenodo.23675104; 10.5281/zenodo.23675278).
- We declare no competing financial or non-financial conflicts of interest.

Thank you very much for your time, consideration, and editorial guidance.

Sincerely,

Martín Sánchez Insfrán, MD
Independent Researcher / Surgeon
Asunción, Republic of Paraguay
ORCID: 0000-0003-4138-9668
Email: `sanchez.martin093@gmail.com`
Code Repository: https://github.com/sanchezmartin093-code/Incendios_PY